

EXP. NO. 7

DATE: 30/09/2025

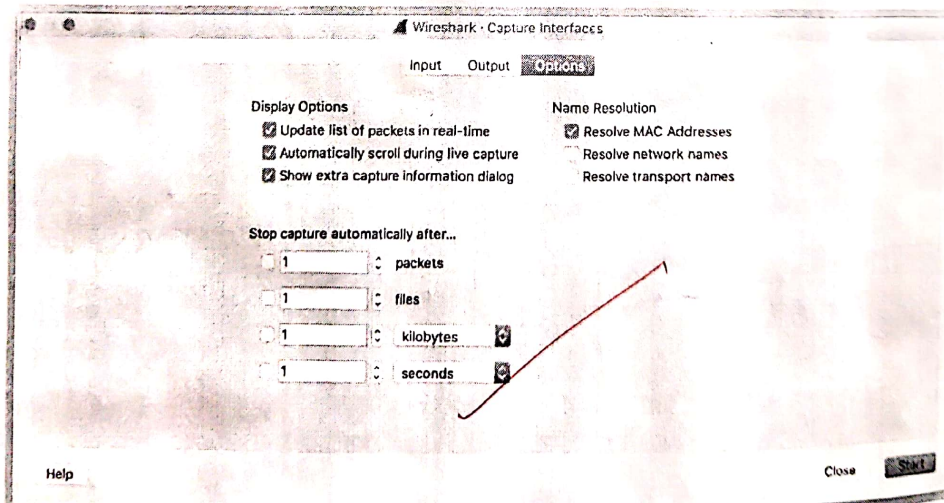
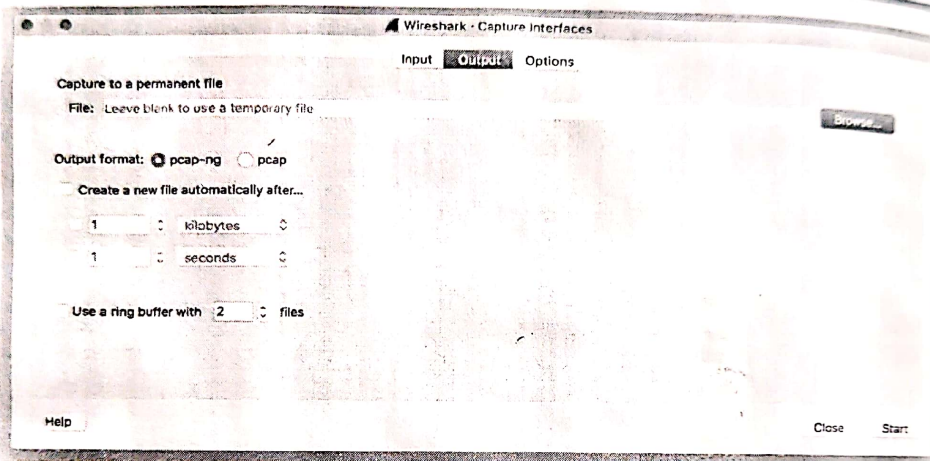
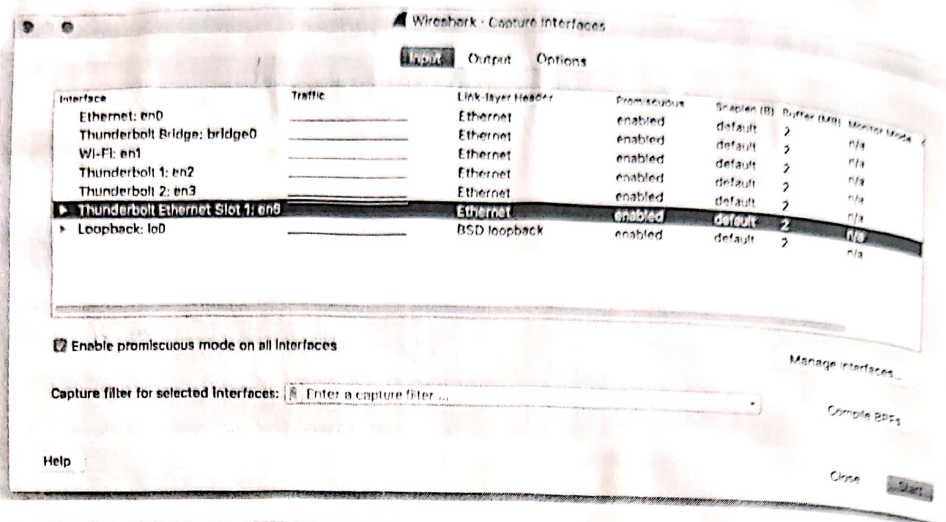
ANALYSIS OF TRAFFIC PACKETS USING WIRESHARK

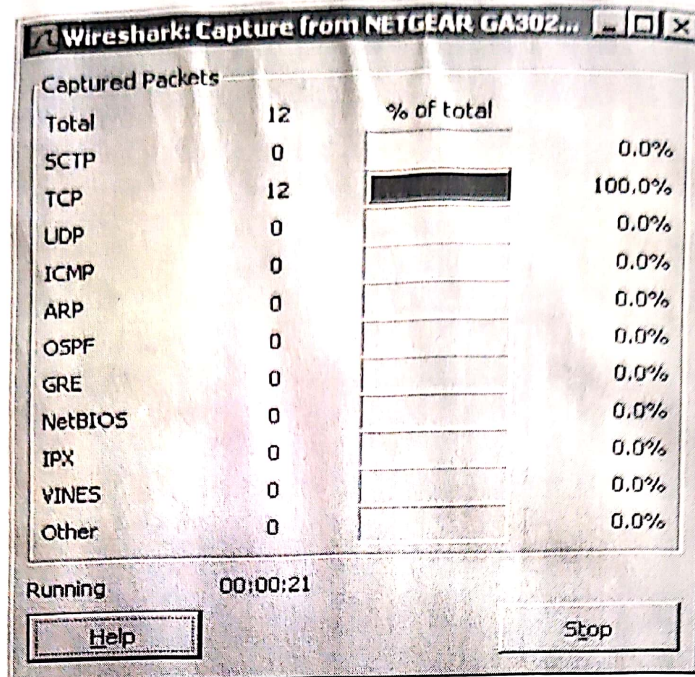
AIM

To analyze network packets using Wireshark, a packet sniffer and analyzer, in order to understand the flow of messages.

PROCEDURE

1. Download and install Wireshark and its library on a computer.
2. Launch the program.
3. Choose the active network interface for packet capture from the Capture menu.
4. Click "Start" to begin capturing network packets visible to the selected interface.
5. Open a web browser and visit a website to create network traffic.
6. After the page loads, stop the packets capture via the Wireshark interface.
7. Use the display filter to view only relevant protocol messages (like HTTP).
8. Click on specific packets to view details of their protocol fields across layers.





Thunderbolt Ethernet Slot 1: en6

Filter: http && (ip.src == 128.2.57.100 || ip.dst == 128.2.57.100)

No.	Time	Source	Destination	Protocol	Length	Info
228	3.499269	128.2.57.100	128.2.131.66	HTTP	716	GET /~ini740/Lab0/lab0.html HTTP/1.1
243	3.616258	128.2.131.66	128.2.57.100	HTTP	748	HTTP/1.1 200 OK (text/html)
250	3.685555	128.2.57.100	128.2.131.66	HTTP	660	GET /favicon.ico HTTP/1.1
252	3.686954	128.2.131.66	128.2.57.100	HTTP	1241	HTTP/1.1 200 OK (Image/x-icon)

Frame 228: 716 bytes on wire (5728 bits), 716 bytes captured (5728 bits) on interface 0

Ethernet II, Src: ElgatoLL_09:48:7d (00:0c:6c:09:48:7d), Dst: Carnegie_0a:00:57 (08:00:7f:0a:00:57)

Internet Protocol Version 4, Src: 128.2.57.100, Dst: 128.2.131.66

Transmission Control Protocol, Src Port: 50696 (50696), Dst Port: 80 (80), Seq: 1, Ack: 1, Len: 650

Hypertext Transfer Protocol

GET /~ini740/Lab0/lab0.html HTTP/1.1

[Expert Info (Chat/Sequence): GET /~ini740/Lab0/lab0.html HTTP/1.1]

[GET /~ini740/Lab0/lab0.html HTTP/1.1]

[Severity level: Chat]

[Group: Sequence]

Request Method: GET

Request URI: /~ini740/Lab0/lab0.html

Request Version: HTTP/1.1

Host: www.ece.cmu.edu

Connection: keep-alive

Upgrade-Insecure-Requests: 1

User-Agent: Mozilla/5.0 (Macintosh; Intel Mac OS X 10_11_5) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/51.0.2704

Accept: text/html,application/xhtml+xml,application/xml;q=0.9,image/webp,*/*;q=0.8

Accept-Encoding: gzip, deflate, sdch

Accept-Language: en-US,en;q=0.8

0020 83 42 c6 00 00 50 b0 13 4d 35 69 0b 8c 9e 80 18 .B...P..M51....

0030 10 15 bf 5b 00 00 01 01 08 0a 1e 2e 1d 30 44 32 ...l....002

0040 e0 a4 47 45 54 20 2f 7e 69 6e 69 37 34 30 2f 4c ..GET /~ ini740/L

0050 61 62 30 2f 6c 61 62 30 2e 68 74 60 6c 20 48 54 ab0/lab0 <html HT

0060 54 50 2f 31 2e 31 0d 0a 48 6f 73 74 3a 20 77 77 TP/1.1. Host: ww

0070 77 2e 65 63 65 2e 63 6d 75 2e 65 64 75 0d 0a 43 w.ece.cm u.edu..C

0080 6f 6e 6e 65 63 74 69 6f 6e 3a 20 6b 65 65 70 2d onnectio n: keep-

0090 61 6c 69 76 65 0d 0a 55 70 67 72 61 64 65 2d 49 alive..U pgrade-I

00a0 6c 73 65 63 75 72 65 2d 52 65 71 75 65 73 74 73 nsecure- Requests

00b0 3a 20 31 0d 0a 55 73 65 72 2d 41 67 65 6e 74 3a : 1..Use r-Agent:

00c0 20 4d 6f 7a 69 6c 6c 61 2f 35 2e 30 20 28 4d 61 Mozilla /5.0 (Mo

00d0 63 69 6e 74 6f 73 60 3b 20 49 6e 74 65 6c 20 4d cintosh; Intel M

00e0 61 63 20 4f 53 20 58 20 31 30 5f 31 31 5f 35 29 ac OS X 10_11_5)

Packet 383: Displayed: 4 (10%) | Dropped: 0 (0.0%) | Profile: Default

9. Try additional filter and display options to analyze the captured data more deeply.
10. Finish your analysis and save the application.

RESULT

By using Wireshark, it is possible to capture, filter, and analyze network traffic, understanding how various protocol messages are structured and exchanged in real-time.